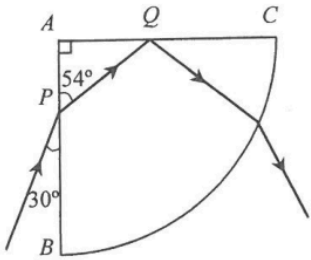
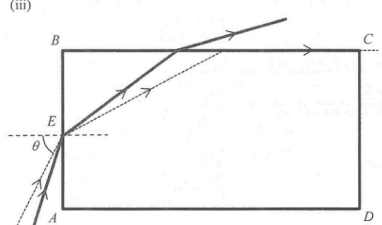
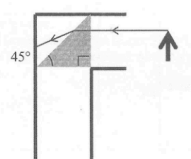
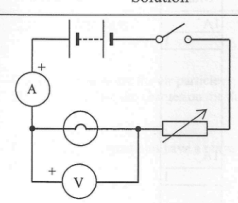


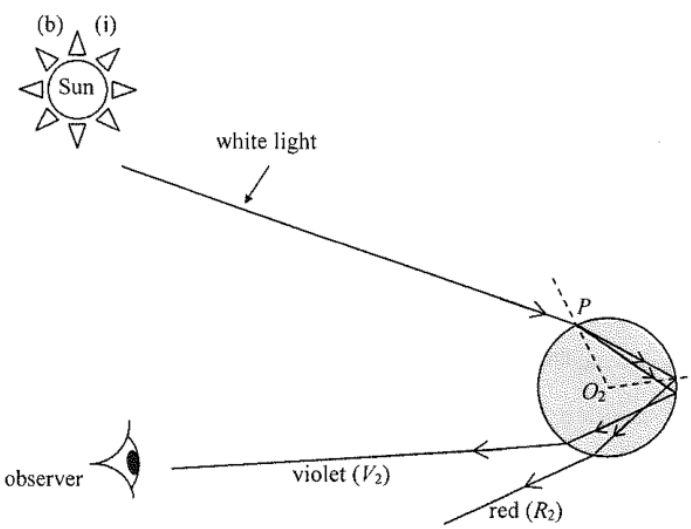
| Solution                                                                                                                                                                                              | Marks             | Remarks             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|---------------------|
| 5. (a) $n = \frac{\sin i}{\sin r}$ (or $n_1 \sin \theta_1 = n_2 \sin \theta_2$ )<br>$= \frac{\sin(90^\circ - 30^\circ)}{\sin(90^\circ - 54^\circ)} = \frac{\sin 60^\circ}{\sin 36^\circ}$<br>$= 1.47$ | 1M<br><br>1A      | Accept 1.47 to 1.50 |
| (b) $\sin c = \frac{1}{n} = \frac{1}{1.47}$<br>$c = 42.7^\circ$ (for $n = 1.50$ , $c = 41.8^\circ$ )<br>Because incident angle at face AC ( $= 54^\circ$ ) $>$ $c$ ( $= 42.7^\circ$ ).                | 2<br><br>1M<br>1M |                     |
| (c) <br><br>Reflected ray correct, $i = r$<br>Emergent ray away from normal.                                         | 1A<br>1A          |                     |
| (d) A spectrum is seen.<br><u>Or</u> splitting into different colours occurs.                                                                                                                         | 1A<br>1A          |                     |
|                                                                                                                                                                                                       | 1                 |                     |

|                                                                                                                                                                                                                                                                                                                                                             |                               |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|--|
| <p>7. (a) (i) At the critical angle <math>c</math>,</p> $\frac{\sin 90^\circ}{\sin c} = n$ $\frac{1}{\sin c} = 1.36$ $c = 47.3^\circ$                                                                                                                                                                                                                       | <p>1M<br/>1A<br/>2</p>        |  |
| <p>(ii) Angle of refraction at <math>E = 90^\circ - 47.33^\circ = 42.67^\circ</math><br/>By Snell's law</p> $\frac{\sin \theta}{\sin 42.67^\circ} = 1.36$ $\theta = 67.2^\circ$                                                                                                                                                                             | <p>1M<br/>1M<br/>1A<br/>3</p> |  |
| <p>(iii)</p>                                                                                                                                                                                                                                                               | <p>2A<br/>2</p>               |  |
| <p>(b) (i)</p>  <p>The angle of incidence of the light ray from the object is (<math>45^\circ</math>, which is) less than the critical angle of the plastic prism.<br/>Total internal reflection will not occur and the image may not be clear enough for observation.</p> | <p>1A<br/>1A<br/>1A<br/>3</p> |  |
| <p>(ii) Glass prism (with critical angle less than <math>45^\circ</math>)<br/><b>Or</b><br/>Plane mirror</p>                                                                                                                                                                                                                                                | <p>1A<br/>1A<br/>1</p>        |  |

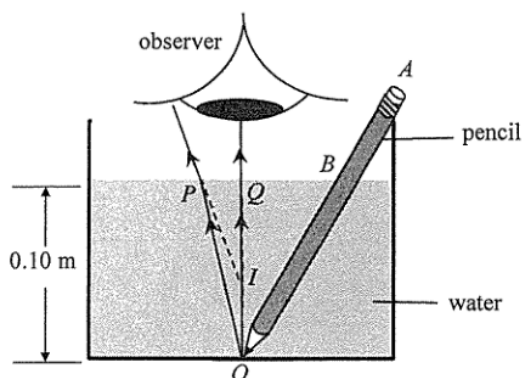
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|                                                                                                                                                                                                                      |                               |                                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>8. (a)</p>                                                                                                                     | <p>1A<br/>1A<br/>1A<br/>3</p> | <p>Correct symbols of light bulb, variable resistor and voltmeter<br/>Correct positions<br/>Correct positive terminal connection for the voltmeter</p> |
| <p>(b) As the voltage across the light bulb increases, the temperature of the light bulb increases, thus its resistance increases.</p>                                                                               | <p>1A<br/>1A<br/>2</p>        |                                                                                                                                                        |
| <p>(c) <math>R = \frac{V}{I}</math> is the definition of resistance.<br/>It is applicable to all conductors.</p>                                                                                                     | <p>1A<br/>1</p>               |                                                                                                                                                        |
| <p>(d) At <math>V = 0.1 \text{ V}</math></p> $R = \frac{V}{I} = \frac{0.1}{76 \times 10^{-3}} = 1.32 \Omega$ <p>At <math>V = 2.5 \text{ V}</math></p> $R = \frac{V}{I} = \frac{2.5}{250 \times 10^{-3}} = 10 \Omega$ | <p>1A<br/>1A<br/>3</p>        | <p>for reading correct values (ignore the order of magnitude) from the graph</p>                                                                       |
| <p>(e)</p> $R = \rho \frac{l}{A}$ $l = \frac{RA}{\rho}$ $= \frac{1.32 \times (1.66 \times 10^{-9})}{5.6 \times 10^{-8}}$ $= 0.039 \text{ m}$                                                                         | <p>1M+1M<br/>1A<br/>3</p>     |                                                                                                                                                        |

|    |     |       |                                                                                                                                                |                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                              |                                                                                                                                                                                                                                                                                                                    |    |
|----|-----|-------|------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 7. | (a) | (i)   | Angle of incidence at $A$ ,<br>$i_A = 90^\circ - 30^\circ = 60^\circ$                                                                          | 1A                                                                                                                                                                                                                                                                                   | <div>1M for correct equation with angle of refraction in the cladding = <math>90^\circ</math></div> <div>Accept 1.2 or <math>\frac{2}{\sqrt{3}}</math></div> <div>Correct spelling</div> <div>Accept <math>\theta \geq 30^\circ</math></div> <div>1A for ‘taking different paths’</div> <div>1A for ‘different arrival time’</div> <div>NOT accept:<br/>Some of the energy (going at angles larger than <math>30^\circ</math> from the axis) would be lost due to leakage, thus lower intensity.</div> <div>0 A for the explanation if the choice for the change of <math>n_c</math> is INCORRECT.</div> <div>Note: More light rays will escape the optical fibre for a larger <math>n_c</math>.</div> |                                                                                                                                              |                                                                                                                                                                                                                                                                                                                    |    |
|    |     |       |                                                                                                                                                | 1                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                              |                                                                                                                                                                                                                                                                                                                    |    |
|    |     | (ii)  | $n_g \sin c = n_c \sin 90^\circ$<br><br>$\Rightarrow \frac{n_g}{n_c} = \frac{1}{\sin c} \geq \frac{1}{\sin 60^\circ} = 1.1547005 \approx 1.15$ | 1M                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                              |                                                                                                                                                                                                                                                                                                                    |    |
|    |     |       |                                                                                                                                                | 1A                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                              |                                                                                                                                                                                                                                                                                                                    |    |
|    |     |       |                                                                                                                                                | 2                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                              |                                                                                                                                                                                                                                                                                                                    |    |
|    |     | (iii) | Total internal reflection.<br>For light rays from $O$ with $\theta > 30^\circ$ ,<br>they will fail to undergo total internal reflection.       | 1A                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                              |                                                                                                                                                                                                                                                                                                                    |    |
|    |     |       |                                                                                                                                                | 1A                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                              |                                                                                                                                                                                                                                                                                                                    |    |
|    |     |       |                                                                                                                                                | 2                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                              |                                                                                                                                                                                                                                                                                                                    |    |
|    |     | (b)   | (i)                                                                                                                                            | Some light / energy (of the narrow light pulse) taking the <u>shortest path</u> (i.e. $OD$ ) <u>arrives first</u> while the rest of the energy taking <u>longer paths</u> arrives later.<br>Therefore the pulse is broader and the height of the pulse is lower (smaller intensity). |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1A                                                                                                                                           |                                                                                                                                                                                                                                                                                                                    |    |
|    |     |       |                                                                                                                                                |                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1A                                                                                                                                           |                                                                                                                                                                                                                                                                                                                    |    |
|    |     |       |                                                                                                                                                |                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | NOT accept:<br>Some of the energy (going at angles larger than $30^\circ$ from the axis) would be lost due to leakage, thus lower intensity. |                                                                                                                                                                                                                                                                                                                    |    |
|    |     |       |                                                                                                                                                |                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2                                                                                                                                            |                                                                                                                                                                                                                                                                                                                    |    |
|    |     |       |                                                                                                                                                |                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | (ii)                                                                                                                                         | The refractive index of cladding should be increased,<br>as $\frac{n_g}{n_c} = \frac{1}{\sin c}$ , by <u>making <math>c / \sin c</math> larger</u> , only those light rays close to the axis / within a smaller $\theta$ are totally internal reflected,<br><br>(such that $\frac{n_g}{n_c}$ getting closer to 1). | 1A |
|    |     |       |                                                                                                                                                |                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                              |                                                                                                                                                                                                                                                                                                                    | 1A |
|    |     | 2     |                                                                                                                                                |                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                              |                                                                                                                                                                                                                                                                                                                    |    |

| Solution                                                                                                                                                                                                           | Marks                     | Remarks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|---------|
| 6. (a) (i) $\sin r = \frac{\sin 45^\circ}{1.325}$<br>$r = 32.253454^\circ \approx 32.3^\circ$                                                                                                                      | 1M<br>1A                  |         |
| (ii) $c = \sin^{-1} \frac{1}{1.325}$<br>$c = 49.000646^\circ \approx 49.0^\circ$<br>$r \text{ (or } \angle OBA) = 32.3^\circ < c = 49.0^\circ$ ,<br>therefore total internal reflection does not occur at point B. | 2<br>1M<br>1A<br>1A<br>3  |         |
| (b) (i)                                                                                                                          | 2A<br>2                   |         |
| (ii) There is more light energy absorbed by water droplet due to a longer path.<br><u>Or</u><br>There is more light energy transmitted to air at the point of reflection as the light undergoes reflection twice.  | 1A<br>1A<br>1A<br>1A<br>2 |         |

8. (a)



(b)

$$1.33 = \frac{\sin r}{\sin 1.5^\circ}$$

$$r = 1.995175^\circ \approx 2.00^\circ$$

(c)

$$\frac{h_1}{h} = \frac{\tan 1.5^\circ}{\tan 2.00^\circ} = 0.751681 \text{ (or } \tan 1.5^\circ = \frac{PQ}{0.10} \text{ \& } \tan 2^\circ = \frac{PQ}{h_1} \text{)}$$

$$\Rightarrow h_1 \approx 0.075 \text{ m (i.e. 7.5 cm)}$$

1A

1M

1M for image  $I$  formed by the two refracted rays.

2

1M

1A

2

1M

1A

e.c.f. from (b), explicit step / method shown

Accept:  $\frac{\text{real depth}}{\text{apparent depth}} = \frac{h}{h_1} = n$ 

2